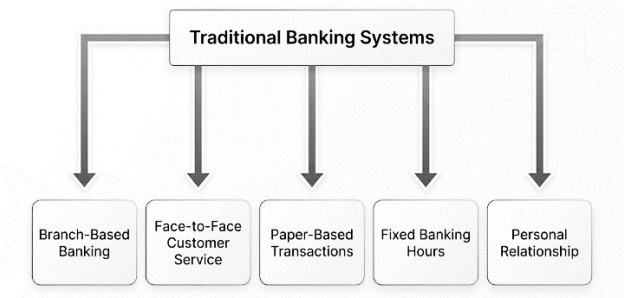
MDM: Financial Technology (S.Y. SEM: III)

Chapter No: 2 Banking and Financial Services

Notes

Q1. Explain the Traditional Banking System. Discuss its features, functions, and importance in the financial sector.

The Traditional Banking System is the conventional method of banking in which customers visit a bank branch to perform banking activities such as depositing money, withdrawing cash, applying for loans, and making payments. It mainly depends on physical branches and face-to-face interactions between customers and bank staff.



Features of the Traditional Banking System

1. *Branch-Based Banking:* Traditional banking mainly operates through physical branches where customers visit the bank to perform financial transactions.
2. *Face-to-Face Customer Service:* Customers interact directly with bank employees for opening accounts, depositing money, applying for loans, and resolving banking issues.
3. *Paper-Based Transactions:* Most banking activities involve physical documents, forms, passbooks, and cheque books.
4. *Fixed Banking Hours:* Banking services are available only during the working hours of the bank branches.
5. *Personal Relationship:* Traditional banking focuses on building long-term relationships between customers and the bank through personalized services.

Functions of the Traditional Banking System

1. Traditional banks accept deposits from individuals, businesses, and other organizations.
2. They provide loans for education, housing, agriculture, business, and personal needs.
3. They offer payment and fund transfer services through cheques, demand drafts, NEFT, RTGS, and IMPS.
4. They provide locker facilities for the safe storage of valuable documents and jewellery.

Importance of the Traditional Banking System

1. Traditional banking provides a safe place for customers to keep their money.
2. It supports economic development by providing loans to individuals and businesses.
3. It builds customer confidence through direct interaction and personalized banking services.
4. It promotes financial discipline by encouraging regular savings.
5. It plays an important role in the growth and stability of the Indian financial system.

Q2. Differentiate between Retail Banking and Corporate Banking with suitable examples.

Basis	Retail Banking	Corporate Banking
Meaning	Retail banking provides banking services to individual customers.	Corporate banking provides banking services to companies, industries, and business organizations.
Customers	Individual customers, salaried employees, students, and pensioners are the main customers.	Companies, partnerships, public sector enterprises, and multinational corporations are the main customers.
Services	Retail banking offers savings accounts, current accounts, personal loans, education loans, home loans, and debit or credit cards.	Corporate banking offers business loans, working capital finance, project finance, trade finance, and cash management services.
Loan Size	The loan amount is generally smaller because it is provided to individuals.	The loan amount is generally larger because it is provided to business organizations.
Risk	The level of risk is usually lower because loans are distributed among many customers.	The level of risk is comparatively higher because large amounts are lent to fewer customers.
Example	SBI provides home loans, education loans, and savings accounts to individuals.	SBI provides project finance and working capital loans to companies such as Tata Motors and Reliance Industries.

Q3. Explain the various Banking Operations and Services provided by Commercial Banks.

Commercial banks perform various banking operations and provide financial services to individuals, businesses, and governments.

Banking Operations and Services

1. *Accepting Deposits:* Commercial banks accept different types of deposits such as savings accounts, current accounts, fixed deposits, and recurring deposits from customers.
2. *Providing Loans:* Banks provide various loans including personal loans, home loans, vehicle loans, education loans, agricultural loans, and business loans according to customer requirements.



3. *Fund Transfer Services:* Banks provide safe and quick transfer of money through NEFT, RTGS, IMPS, UPI, demand drafts, and cheque facilities.
4. *Digital Banking Services:* Commercial banks offer internet banking, mobile banking, ATM services, debit cards, credit cards, and QR code-based payment services.
5. *Locker Facilities:* Banks provide lockers where customers can safely store jewellery, important documents, and other valuable items.
6. *Foreign Exchange Services:* Banks facilitate foreign currency exchange, international money transfers, and foreign trade transactions.
7. *Investment and Insurance Services:* Many commercial banks provide mutual fund investments, insurance products, fixed deposits, and pension schemes.
8. *Collection and Payment Services:* Banks collect cheques, dividends, utility bills, taxes, and insurance premiums on behalf of customers.

Q4. Discuss the role of banks in the Indian financial system. Explain the major functions performed by banks.

Banks are one of the most important financial institutions in India. They act as intermediaries between savers and borrowers by collecting deposits from the public and providing loans to individuals, businesses, and governments. Banks contribute significantly to economic growth and financial stability.

Role of Banks in the Indian Financial System

1. Banks encourage people to save money by offering different types of deposit accounts and investment schemes.
2. They provide financial assistance to industries, agriculture, businesses, and individuals through different types of loans.
3. Banks support economic development by financing infrastructure projects and industrial expansion.
4. They promote financial inclusion by providing banking services in rural and remote areas.

5. Banks facilitate digital payments and electronic banking services, making financial transactions faster and more secure.
6. They help implement government schemes and welfare programs by transferring benefits directly to beneficiaries.

Major Functions of Banks

1. *Accepting Deposits:* Banks collect savings from the public through savings, current, fixed, and recurring deposit accounts.
2. *Providing Credit:* Banks provide loans for housing, education, agriculture, business, and personal requirements.
3. *Creating Credit:* Banks create credit by lending a portion of the money received as deposits.
4. *Payment and Settlement Services:* Banks provide payment services through UPI, NEFT, RTGS, IMPS, debit cards, and cheque clearing.
5. *Agency Services:* Banks collect bills, dividends, insurance premiums, and taxes on behalf of customers.
6. *General Utility Services:* Banks provide lockers, foreign exchange services, investment guidance, and financial advisory services.

Q5. What is Financial Inclusion? Explain its objectives, benefits, challenges, and initiatives taken in India.

Financial Inclusion is the process of providing affordable and accessible financial services to all sections of society, especially low-income groups, rural populations, women, and economically weaker sections. These services include savings accounts, loans, insurance, pension schemes, and digital payment facilities.



Objectives of Financial Inclusion

1. Financial inclusion aims to provide banking and financial services to every citizen at an affordable cost.
2. It encourages people to save money through formal banking institutions instead of informal sources.
3. It improves access to credit for small businesses, farmers, and economically weaker sections.
4. It promotes digital payments and reduces dependence on cash transactions.
5. It supports inclusive economic growth by ensuring equal access to financial services.

Benefits of Financial Inclusion

1. Financial inclusion improves the standard of living by providing access to banking and credit facilities.
2. It encourages savings and promotes financial security among individuals and families.
3. It reduces dependence on moneylenders who often charge very high interest rates.
4. It supports entrepreneurship by providing loans to small businesses and self-employed individuals.
5. It increases transparency in government benefit transfers through Direct Benefit Transfer (DBT).

Challenges of Financial Inclusion

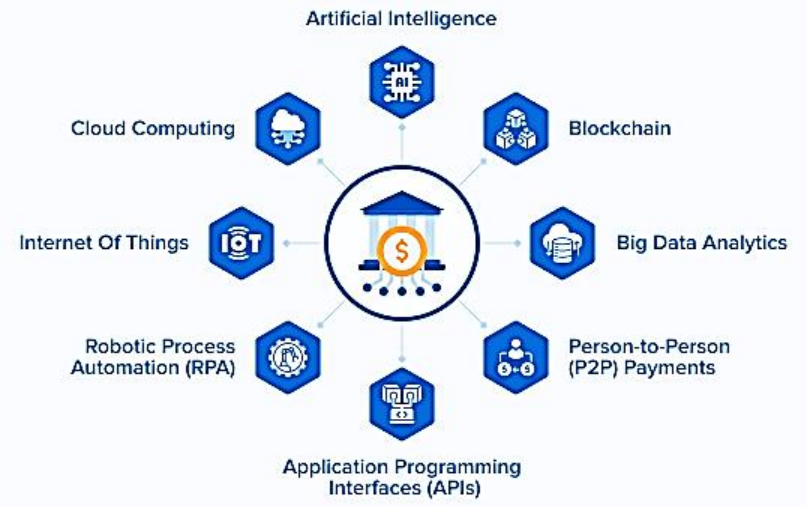
1. Many rural and remote areas still have limited access to banking facilities.
2. Lack of financial literacy prevents many people from using banking services effectively.
3. Poor internet connectivity affects the use of digital banking in some regions.
4. Low income and irregular employment reduce the ability of people to save and use financial products.
5. Cybersecurity risks and online fraud discourage some people from using digital financial services.

Major Initiatives Taken in India

1. *Pradhan Mantri Jan Dhan Yojana (PMJDY):* This scheme provides basic bank accounts with debit cards, insurance benefits, and overdraft facilities to promote financial inclusion.
2. *Digital India Initiative:* This initiative encourages digital payments and online banking services across the country.
3. *Unified Payments Interface (UPI):* UPI enables secure and instant money transfers using mobile phones.
4. *Direct Benefit Transfer (DBT):* Under this scheme, government subsidies and welfare benefits are directly transferred to the bank accounts of beneficiaries.
5. *Pradhan Mantri Mudra Yojana (PMMY):* This scheme provides loans to small businesses and entrepreneurs without requiring collateral.

Q6. Explain how Digital Transformation has changed the banking sector. Discuss its advantages and challenges.

Digital Transformation in banking refers to the use of digital technologies to provide faster, safer, and more convenient banking services. It has changed traditional banking by introducing online and mobile-based financial services.



Changes Brought by Digital Transformation

1. Banking services are now available online through websites and mobile applications. Customers can access their accounts and perform transactions without visiting a bank branch.
2. Digital payment systems have become widely used. Services such as UPI, NEFT, RTGS, IMPS, and mobile wallets have made money transfers faster and more secure.
3. Banks provide 24×7 banking services. Customers can access banking facilities at any time and from any location.
4. Paperless banking has reduced the use of physical documents. Many banking processes such as account opening, loan applications, and KYC verification are completed digitally.
5. Banks use technologies such as Artificial Intelligence (AI), cloud computing, and data analytics. These technologies help banks improve customer service and detect fraudulent transactions.

Advantages of Digital Transformation

1. Digital banking provides quick and convenient banking services without requiring customers to visit a bank branch.
2. It reduces the time required for transactions such as fund transfers, bill payments, and account management.
3. It lowers the operating cost of banks by reducing paperwork and manual processing.
4. It improves customer satisfaction by providing personalized and faster services.
5. It promotes financial inclusion by providing banking services in rural and remote areas through digital platforms.

Challenges of Digital Transformation

1. Cybersecurity threats such as hacking, phishing, and online fraud have increased.
2. Some customers have limited digital literacy and face difficulty in using online banking services.
3. Poor internet connectivity affects digital banking services in some rural areas.
4. Technical failures and server downtime may temporarily interrupt banking services.

Q7. Describe the Evolution of Electronic Banking. Explain the different stages of development from traditional banking to digital banking.

Electronic Banking (E-Banking) is the use of electronic technologies to provide banking services through computers, mobile phones, ATMs, and the internet. It has evolved over several years with the advancement of information and communication technology.

Stages of Evolution

1. Traditional Banking

In the early stage, customers visited bank branches for all banking activities such as deposits, withdrawals, and loan applications. Most transactions were carried out manually using paper records.

2. Computerized Banking

Banks introduced computers to maintain customer records, process transactions, and improve operational efficiency. Although banking became faster, customers still depended mainly on physical branches.

3. ATM Banking

Banks introduced Automated Teller Machines (ATMs), allowing customers to withdraw cash, check account balances, and perform basic banking services without visiting a bank counter.

4. Internet Banking

Banks launched internet banking services, enabling customers to access their accounts, transfer funds, pay bills, and download account statements using computers connected to the internet.

5. Mobile Banking

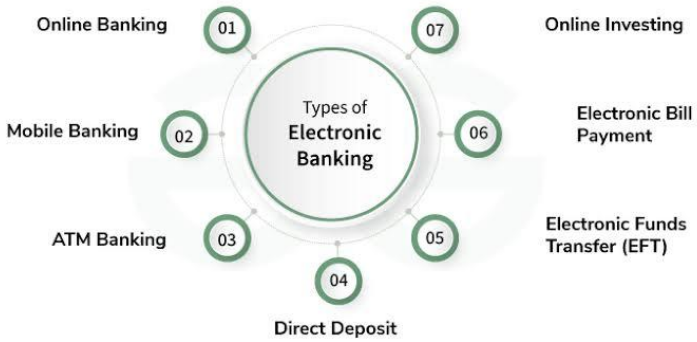
The development of smartphones allowed banks to provide mobile banking applications. Customers could perform banking transactions anytime and anywhere using their mobile devices.

6. Digital Banking

Digital banking combines internet banking, mobile banking, UPI, artificial intelligence, cloud computing, and digital payment systems to provide complete banking services without requiring physical branch visits.

Q8. Explain various Electronic Banking (E-Banking) services such as Internet Banking, Mobile Banking, ATM, UPI, NEFT, RTGS, and IMPS.

Electronic Banking (E-Banking) allows customers to access banking services electronically without visiting a bank branch.



Types of E-Banking Services

1. Internet Banking

Internet banking allows customers to access their bank accounts through a secure website using a computer or mobile device. Customers can transfer money, pay bills, check balances, and download account statements.

2. Mobile Banking

Mobile banking enables customers to perform banking transactions through a bank's mobile application. It provides services such as fund transfers, account management, utility bill payments, and mobile recharges.

3. Automated Teller Machine (ATM)

An ATM allows customers to withdraw cash, deposit money, check account balances, change PINs, and obtain mini statements without visiting a bank branch.

4. Unified Payments Interface (UPI)

UPI is a real-time digital payment system that enables instant transfer of money between bank accounts using mobile applications. It operates 24 hours a day and requires only a mobile number or UPI ID.

5. National Electronic Funds Transfer (NEFT)

NEFT is an electronic fund transfer system that allows customers to transfer money from one bank account to another anywhere in India. It is suitable for small and medium-value transactions.

6. Real Time Gross Settlement (RTGS)

RTGS is used for transferring large amounts of money instantly. Transactions are settled individually in real time, making it suitable for high-value payments.

7. Immediate Payment Service (IMPS)

IMPS is an instant electronic fund transfer service that allows customers to transfer money at any time, including weekends and holidays. It is commonly used for urgent transactions.

Q9. Explain the relationship between Financial Inclusion and Digital Transformation. How do they contribute to inclusive economic growth?

Financial Inclusion and Digital Transformation are closely connected. Digital technology makes banking services more accessible, while financial inclusion ensures that these services reach every section of society.

Relationship between Financial Inclusion and Digital Transformation

1. Digital transformation enables people to access banking services through mobile phones and the internet, reducing the need to visit bank branches.
2. Financial inclusion ensures that banking services are available to rural populations, low-income groups, women, and small businesses through digital platforms.
3. Digital payment systems such as UPI, mobile banking, and internet banking make financial services affordable, fast, and accessible to everyone.
4. Government schemes such as Pradhan Mantri Jan Dhan Yojana (PMJDY) and Direct Benefit Transfer (DBT) use digital banking to deliver financial benefits directly to citizens.

Contribution to Inclusive Economic Growth

1. Financial inclusion encourages more people to participate in the formal banking system.
2. Digital banking provides easy access to savings, loans, insurance, and investment services.
3. Small businesses and entrepreneurs can obtain financial support and digital payment facilities to expand their businesses.
4. Digital transactions increase transparency and reduce the use of cash, helping to improve financial accountability.
5. Together, financial inclusion and digital transformation increase employment opportunities, improve financial security, and support sustainable economic growth.

Q10. Explain the impact of Digital Banking and Electronic Banking on customers, banks, and the Indian economy.

Digital Banking and Electronic Banking have significantly improved the efficiency of banking services and transformed the way financial transactions are carried out.

Impact on Customers

1. Customers can access banking services anytime and from any location using mobile phones and computers.
2. Digital banking enables quick fund transfers, online bill payments, shopping, and account management without visiting bank branches.
3. Customers save time and travel costs because most banking services are available online.
4. Digital banking provides secure payment options through UPI, debit cards, internet banking, and mobile banking applications.

Impact on Banks

1. Banks have reduced operational costs by automating many banking processes and reducing paperwork.
2. Digital technologies have improved customer service by providing faster and more personalized banking experiences.
3. Banks use artificial intelligence, data analytics, and cybersecurity systems to improve efficiency and detect fraudulent transactions.
4. Banks can serve a larger number of customers without opening additional physical branches.

Impact on the Indian Economy

1. Digital banking has promoted financial inclusion by extending banking services to rural and underserved areas.
2. It has increased the use of digital payments, reducing dependence on cash transactions.
3. Faster financial transactions have improved business efficiency and supported economic growth.
4. Digital banking has increased transparency in financial transactions and helped reduce tax evasion and corruption.
5. It has supported government initiatives such as Digital India, UPI, and Direct Benefit Transfer (DBT), contributing to a more efficient and inclusive financial system.

Question 11: How does Banks makes money

Net Interest Income

This is the primary way traditional commercial banks generate revenue.

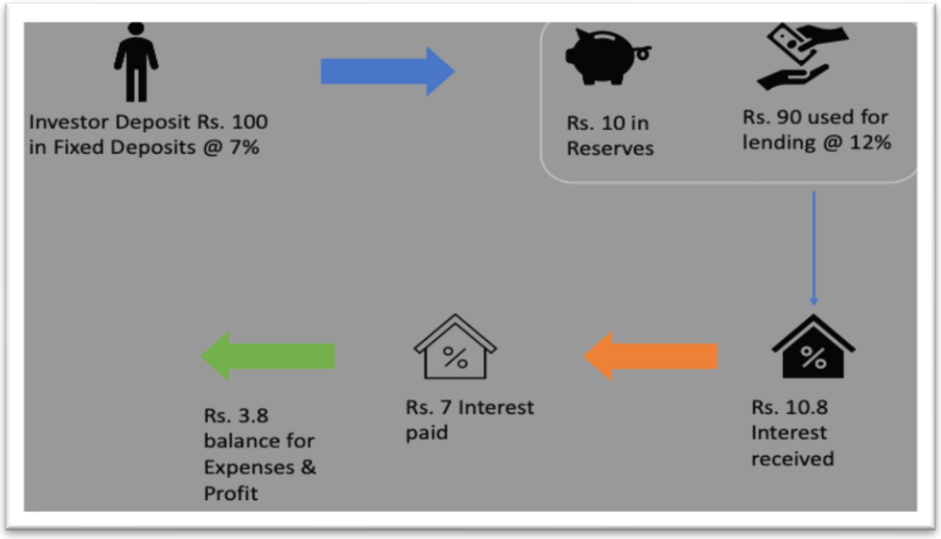
- **Deposits as Cheap Capital:** Banks gather funds from individuals and businesses via savings accounts, current accounts, and fixed deposits. They pay depositors a relatively low interest rate (e.g., 2% to 4%).
- **Lending at Higher Rates:** The bank pools these deposits and lends them out to individuals and corporations through mortgages, car loans, personal loans, and business credits. They charge borrowers a much higher interest rate (e.g., 8% to 12%).
- **Net Interest Margin:** The gap between the interest earned on loans and the interest paid to depositors is the bank's gross profit margin, technically called the **Net Interest Margin (NIM)**.

Other Income

- **Fee-Based & Non-Interest Income**
  - **Transaction & Account Fees:** Charges for falling below minimum monthly balances, processing fund transfers
  - **Wealth Management & Advisory:** Banks earn commissions and brokerage fees by selling third-party insurance policies, mutual funds, portfolio management services, and locker rentals.
  - **Foreign Exchange (Forex):** Banks charge retail markup margins and handling fees when converting currencies for travellers or businesses engaged in international trade.

- **Capital Markets & Investment Banking**
  - **Treasury Operations:** Banks invest excess capital into low-risk government bonds and securities, earning consistent interest coupon payments.
  - **Investment Banking Fees:** Large corporate banks assist companies with initial public offerings (IPOs), underwriting debt, and arranging mergers and acquisitions (M&A), collecting massive advisory fees in the process.

Net Interest Income Schematic:



Question 12: Calculate the Profit earned by Bank during the year

- Bank has a Savings Accounts Deposits @ 4% worth Rs. 10 Crores
- Bank has a Fixed Deposits Accounts Deposits @ 7% worth Rs. 15 Crore
- Bank gave a loan @ 12% of Rs. 20 Crore
- Requirement of Cash Reserve is 10%
- Employee Cost & Establishment Cost of the Bank Rs. 50 lakh for the year
- RBI has penalized the Bank for Rs. 25 lakh during the year for violating KYC regulations
- Bank has earned Capital gain of Rs. 50 Lakh during the year

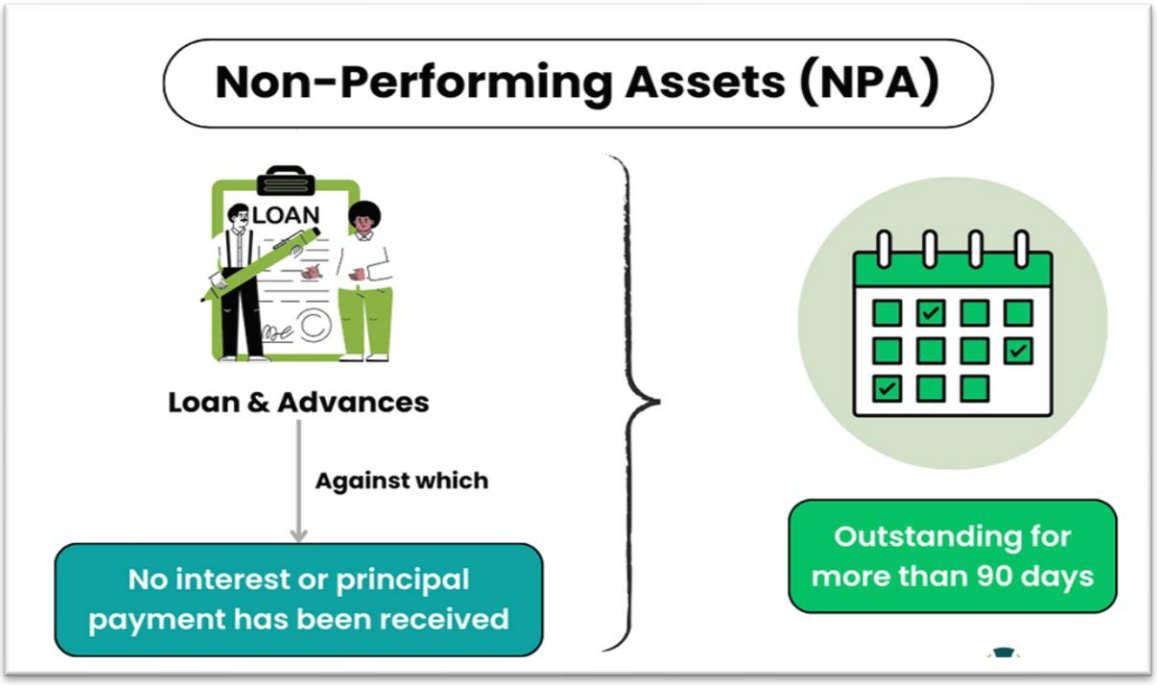
Answer :

Particulars	Principal Amount	Interest Rate	Amount
Interest on Loan	20,00,00,000	12%	2,40,00,000
Capital gain			50,00,000
Total for Income			**2,90,00,000**
Interest on Savings Deposits Account	10,00,00,000	4%	40,00,000
Interest on Fixed Deposits Account	15,00,00,000	7%	1,05,00,000
Employee Cost & Establishment Cost			50,00,000
Penalty from RBI			25,00,000
Total for Expenses			**2,20,00,000**
Profit			**70,00,000**

Question 13: What is NPA & why NPA is bad for the economy ?

NPA

- A Non-Performing Asset (NPA) is simply a loan given by the bank (asset for the bank) that has stopped performing—meaning the borrower has stopped paying the interest or principal back to the bank for a continuous period of 90 days (3 months)



Why NPA is bad for the economy ?

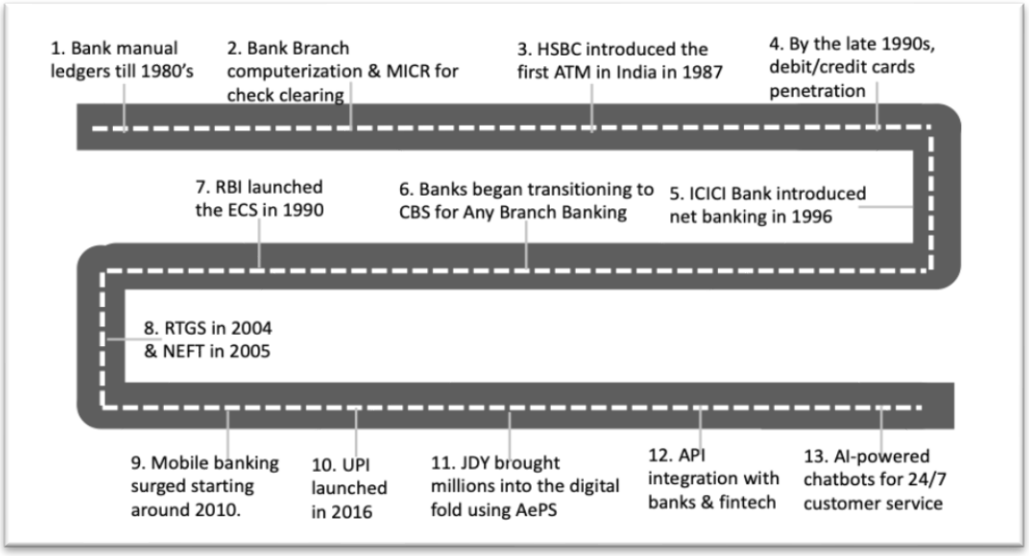
- Under RBI rules, when a loan turns into an NPA, the bank must take a portion of its own profits and lock it away in an emergency fund called Provisions. This means the bank has less profit now.
- To recover the losses caused by defaulters, banks often lower the interest rates they give to savers and increase the loan interest rates for honest borrowers.
- If NPA is high, Banks it runs out of cash to give new loans / credit to genuine people.

Question 14: How did evolution of Electronic Banking take place in India ?

The evolution of electronic banking (E-banking) in India completely transformed the traditional brick-and-mortar banking system into a 24/7 digital ecosystem. This transition took place through a structured, phased approach over the last four decades, heavily guided by the **Reserve Bank of India (RBI)** and technological adoption.

- Bank used to record Banking Transactions (withdrawal, receipt etc.) through manual ledgers.
- **Various RBI's Committees:** RBI-led committees laid the blueprint for bank computerization as well as usage of MICR technology for scanning the cheques.
- **Introduction of ATMs:** In 1987, **HSBC** installed India's first Automated Teller Machine (ATM) in Mumbai, introducing customers to self-service banking.

- **Shared Networks:** In 1997, the Indian Banks' Association (IBA) launched shared network of ATMs, allowing customers to use other banks' machines.
- **Core Banking Solutions (CBS):** Banks migrated from localized servers to centralized databases. This meant a customer was now a "customer of the bank" rather than just a "customer of a specific branch."
- **Electronic Clearing Service (ECS)** : ECS ,an electronic mode of fund transfer was introduced by the Reserve Bank of India (RBI) to handle repetitive, bulk, and periodic payments. Instead of writing individual cheques or handling physical cash, ECS allows institutions and individuals to automate these transactions digitally.
- **Debit / Credit Cards :** By 2000, credit cards as well as debit cards became a popular medium for making the payments.
- **Payment Gateways (RTGS & NEFT):** The RBI introduced **Real Time Gross Settlement (RTGS)** in 2004 for high-value transfers, followed by **National Electronic Funds Transfer (NEFT)** in 2005, replacing slow physical clearinghouses.
- **Cheque Truncation System (CTS):** Launched in 2008, CTS stopped the physical movement of paper cheques, instead transmitting electronic images for faster clearing
- **Immediate Payment Service (IMPS):** Launched in 2010, IMPS provided Indians with their first 24/7/365 instant interbank electronic fund transfer system on mobile phones.
- **Unified Payments Interface (UPI):** Launched in 2016, UPI revolutionized banking by allowing instant, peer-to-peer bank transfers using a simple Virtual Payment Address (VPA) or QR code, eliminating the need for complex IFSC codes and bank account numbers.
- **Financial Inclusion & Micro ATMs:** The launch of the **Aadhaar Enabled Payment System (AePS)** and handheld Micro ATMs allowed business correspondents to bring electronic banking directly to doorstep levels in rural India.
- **Rise of Neo banks :** Post 2020s, Neo-banks (Digital-first bank) like Airtel Payments Bank, Paytm etc, started reshaping access to finance



Question 15: Explain what is KYC , eKYC and CKYC ?

KYC (Know Your Customer)

KYC is a mandatory verification process that banks and financial institutions use to confirm the true identity and address of their customers. Introduced to prevent financial crimes like identity theft, money laundering, and terrorist financing, it ensures that financial platforms are not used for illegal activities.

Whenever you open a bank account, buy a mutual fund, or get a credit card, you must provide proof of who you are.

- **How it works:**
  - You fill out a physical or digital form and submit standard identity proofs (like a PAN card) along with address proofs (like a Passport, Voter ID, or Aadhaar card).
  - Bank / Financial institution verifies the original physical documents against the photocopies you provided.

eKYC (Electronic KYC)

eKYC is a completely paperless, electronic version of traditional KYC that relies on India's national biometric identity database, **Aadhaar**.

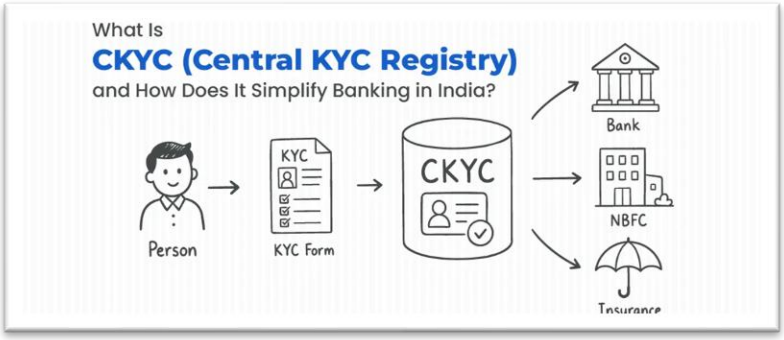
  - Instead of submitting physical photocopies, you share your Aadhaar number. You then verify your identity instantly using either your **biometrics** (fingerprint/iris scan at a branch) or an **OTP** sent to your Aadhaar-registered mobile phone.
  - The system instantly pulls your digitally signed photograph, name, and address directly from the UIDAI database. It reduces verification times from days to a few seconds.

CKYC (Central KYC)

CKYC is a centralized repository designed to eliminate repetitive paperwork across the entire financial sector.

The Problem it Solved: Previously, if you did KYC for Bank A, you had to do it all over again if you opened an account with Mutual Fund B or Stockbroker C.

  - When you complete your verification for the first time at any regulated financial entity, they upload your details to the Central KYC registry. You are issued a unique **14-digit KIN (KYC Identification Number)**.
  - The next time you visit a new financial institution (e.g., buying insurance), you simply give them your KIN. They pull your pre-verified records from the central registry, completely bypassing the need for fresh documentation.



Question 16: Design a Financial Inclusion Plan with a timeline of 1 year for :

- A remote village in rural India has 500 households.
- The nearest commercial bank branch is 15 kilometers away.
- Most villagers are daily-wage farm workers, shopkeepers or artisans.
- They currently rely on informal local moneylenders who charge high interest rates
- They do not use digital payments due to a lack of awareness and connectivity.

To lift the 500 households of this remote village out of the debt trap set by informal moneylenders, following plan is deployed :

1. Foundation & Institutional Access (Months 1–4)

- **Establish a Customer Service Point (CSP)**: Partner with a public sector bank (like State Bank of India) to appoint a local educated youth as a permanent Bank Mitra (Business Correspondent) equipped with a micro-ATM for cash-in/cash-out services, eliminating the 15-km travel barrier.
- **100% Jan Dhan Saturation**: Conduct a weekend camp to open Pradhan Mantri Jan Dhan Yojana(PMJDY) zero-balance accounts for all unbanked adults, bundled with Rupay debit cards and default access to accident insurance.

2. Credit & Debt Restructuring (Months 4–8)

- **Moneylender Buyout Scheme**: Facilitate institutional bank loans through Kisan Credit Card (KCC) for farmers and Mudra loans (Shishu tier up to ₹50,000) for artisans and shopkeepers to aggressively pay off and close legacy high-cost informal loans.
- **Women-Led Self-Help Groups (SHGs)**: Form 10 to 15 Self-Help Groups under the National Rural Livelihoods Mission (NRLM), promoting micro-savings of ₹100–500 per month per member to build an internal emergency lending corpus.

3. Digital Literacy & Connectivity (Months 9–12)

- **Offline and Feature Phone Payments**: Deploy UPI123Pay that enables digital payments on basic 2G feature phones, bypassing the need for smartphones or continuous high-speed data.
- **Nukkad Natak & Visual Training**: Run weekly vernacular street plays and audio-visual demonstrations at weekly haats (markets) illustrating PIN security, voice-prompt transactions, and fraud prevention.

